

Revit for Structures

Course outline steel & concrete structural modeling in Autodesk Revit

REINFORCED CONCRETE TRACK **STEEL TRACK**

PROGRAM OVERVIEW

Revit for Structures is a hands-on Autodesk Revit training program for engineering students and early-career professionals who want to build real, buildable structural models – not just pretty renders. The program runs as two focused tracks that can be taken separately or together: Reinforced Concrete Structures and Steel Structures. Both tracks move from the fundamentals of the Revit interface through full structural modeling, reinforcement/connection detailing, schedules, and construction documentation.

START DATE

Monday, 14 September 2026 third week of September

FORMAT

100% online live instructor-led sessions via Zoom

DURATION

4 weeks per track weekday evenings

FEE

KSH 2,500 per track

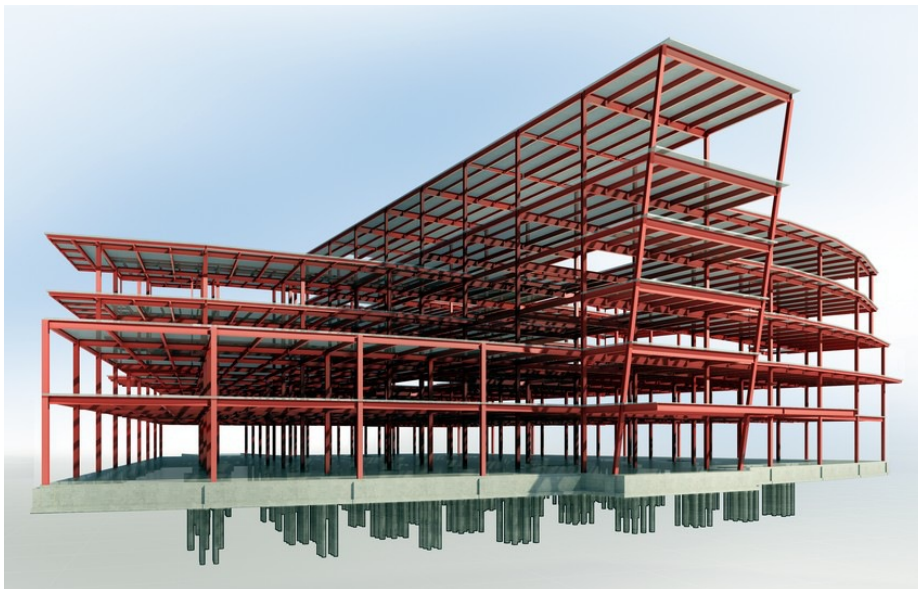
WHO IT'S FOR

University engineering students and early-career structural / BIM professionals

PREREQUISITES

A laptop capable of running Revit; no prior Revit experience required

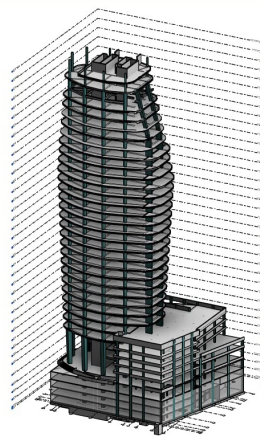
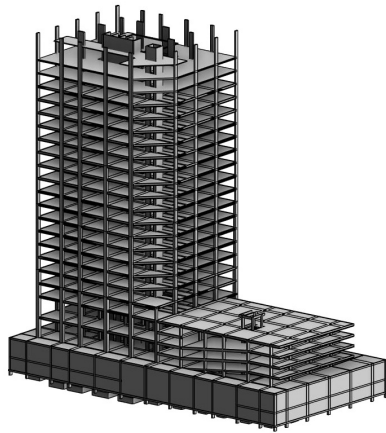
Program contact: structratraining.co.ke +254 7XX XXX XXX



TRACK 01

Reinforced Concrete Structures Track

Modules 1 20 72 hands-on sub-topics datum setup through reinforcement, schedules and documentation



1. Brief overview of Revit

- Difference with AutoCAD software
- Benefits over AutoCAD software

2. Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

3. Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

4. Creating datum elements

- Creating levels
- Creating grids
- Propagating grid extents
- Customizing grid bubble shape and appearance
- Linking CAD files to create grids

5. Overview of creating 3D elements

- Foundations, columns, beams and slabs

6. Creating columns

- Placement of columns
- Loading column families

7. Creating foundations

- Hosting foundations on the columns created
- Creating strip footing
- Using the eccentricity option
- Basement floor slab
- Loading foundation families into your drawings

8. Creating beams and slabs

- Copying beams and slabs from level to level

Creating a slab
Hollow pot and waffle slabs

9. Curved beams

Procedure for modeling curved beams

10. Beam system layout and justification rules

Using the beam system to model structural beams
Modeling beams using beam system layout rules
Modeling beams using beam system justification rules
Deselecting the beam system

11. Creating staircases and ramps

Types of staircases
Creating an architectural ramp and a structural ramp
Modeling a staircase as a ramp
Shaft openings

12. Visibility/graphics and filters

Applying visibility/graphic overrides method 1
Applying visibility/graphic overrides method 2
Applying visibility/graphic overrides method 3
Applying visibility/graphic overrides method 4

13. Creating roofs and truss placement

Sloped and gable-ended roofing
Adding system trusses to your roof
Editing and creating customized trusses

14. Placing reinforcement

Creating sections in your drawing
Foundation reinforcement
Column reinforcement
Beam reinforcement
Slab reinforcement
Staircase reinforcement
Ramp reinforcement

15. Annotations

Adding dimensions
Adding spot elevations
Adding spot coordinates
Adding spot slope

16. Detailing of sections

Placing breaklines in sections
Using hatch files to create regions
Placement of rebar tags on reinforcement
Adding comments on reinforcement
Tagging structural elements

17. Schedules and material takeoffs

Creating bar bending schedules
Loading shape images into bar bending schedules
Creating material takeoffs for structural elements

Sorting and filtering fields in schedules

18. Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template
- Duplicate options for drawings
- Placing drawings on sheets

19. Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

20. Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis

TRACK 02

Steel Structures Track

Modules 1 29 101 hands-on sub-topics datum setup through steel connections, fabrication elements and documentation

1. Brief overview of Revit

- Difference with CAD software
- Benefits over CAD software

2. Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

3. Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

4. Creating datum elements

- Creating levels
- Creating grids
- Elevation markers
- Propagating grid extents
- Customizing grid bubble shape and appearance (optional)
- Linking CAD files to create grids

5. Modeling steel columns

- Placement of columns
- Loading column families

6. Modeling concrete columns

- Placement of concrete columns
- Loading column families

7. Modeling the foundations

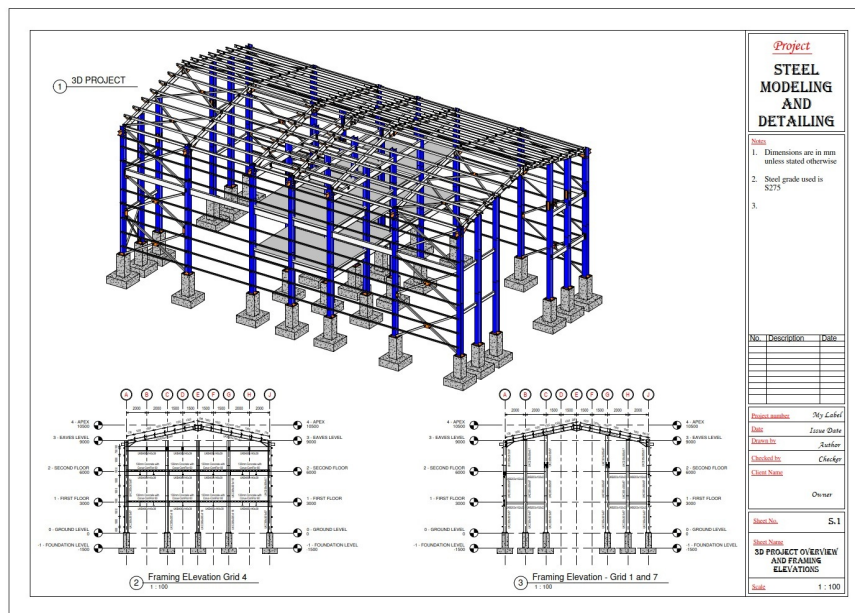
- Hosting foundations on the columns created
- Loading other foundation families into your drawings

8. Modeling the rafters

- Placement of rafters
- Loading steel beam sections
- Attaching columns to the rafters
- Copying rafters from grid to grid

9. Modeling the steel beams

- Loading steel beam sections
- Modeling the floor beams
- Justification rule of the beams
- Copying the floor beams
- Modeling beams at the eaves level



10. Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system

11. Composite slab and composite beam

- Modeling the composite beam
- Modeling the composite slab
- Cantilevering the composite slab

12. Floor openings

Using the shaft opening tool

13. Curved beams

Procedure for modeling curved beams

14. Sloped / slanted beams

Modeling sloped beams using the offset tool

Modeling sloped beams using the framing elevation

15. Purlins and side rails

- Modeling purlins using reference planes
- Modeling purlins using the beam system
- Modeling the side rails

16. Bracings

Vertical bracings
Horizontal bracing

17. Visibility/graphics and filters

Applying visibility/graphic overrides	method 1
Applying visibility/graphic overrides	method 2
Applying visibility/graphic overrides	method 3
Applying visibility/graphic overrides	method 4
Hiding elements	

18. Creating sections

Types of sections

Creating sections in your drawing

19. Placing reinforcement

Setting up the reinforcement cover

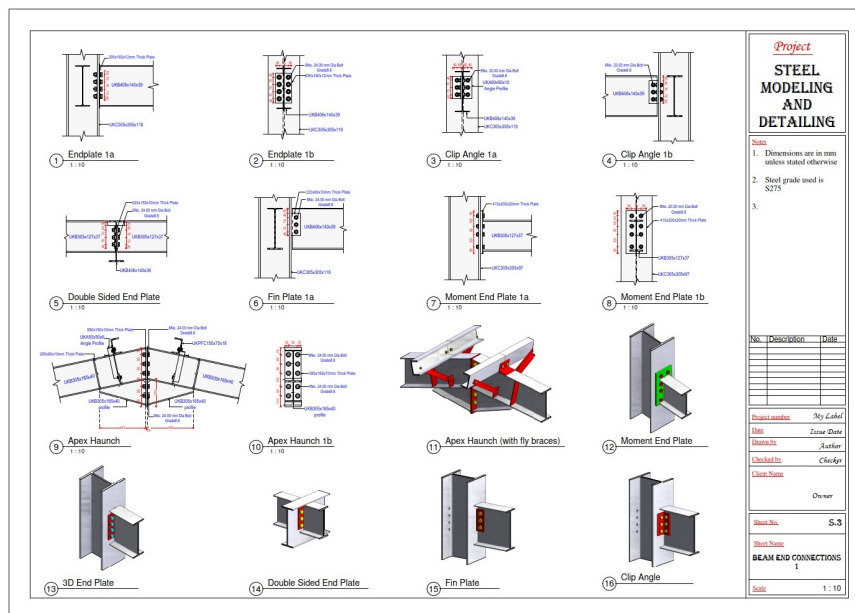
Placing reinforcement at the foundation

Column reinforcement

Slab reinforcement

20. Splitting elements

Splitting elements



21. Introduction to connections

Uploading the connections

Applying connections to structural elements

Propagating the connections

22. Connection types

Main types of connection

Haunch connection

Apex connection

Base plate connection

Gusset plate bracing connection

Gusset plate column-base plate connection

Purlin / side rail connection

Single-sided end plate (simple connection)

Single-sided end plate (moment connection)

Double-sided end plate (simple connection)

Fin plate (simple connection)

Clip angle (simple connection)

Column splice connection

Creating custom connections

23. Fabrication elements, modifiers and parametric cuts

Use and location

24. Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope
- Creating custom bolt and plate tags

25. Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

26. Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

27. Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template
- Duplicate options for drawings
- Placing drawings on sheets

28. Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

29. Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis